REVISION 2026

# Basic Surveying Lab

Diploma in Civil Engineering

Code: 2019 | Credits: 1.5 | Semester: 2

## Course Outcomes (CO)

- **CO1:** Perform linear and angular measurements using chain and compass. (Apply)
- **CO2:** Determine level differences using simple and differential levelling. (Apply)
- **CO3:** Plot longitudinal profiles and prepare contour maps. (Apply)
- **CO4:** Measure horizontal angles and bearings using a theodolite. (Apply)

**Common mistakes and precaution:** Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

## Module 1: Chain & Compass Survey

### Expt 1: Equipment Introduction

Demonstration of basic survey equipment: Chain, Cross-staff, Plane table, and Compass.

### Expt 3: Compass Survey

Determine inaccessible distances between points using compass and tape measurements.

> [illegible]

## Module 2: Levelling

### Expt 4: Simple Levelling

Determine level difference between two points using a Dumpy level.

### Expt 5: Rise and Fall Method

Determine reduced levels (RL) by the Rise and Fall method for a series of points.

> [illegible] *(RL)* [illegible]

## Module 3: Profile & Contouring

### Expt 7: Profile Levelling

Plot longitudinal profiles and cross sections for the alignment of a road.

### Expt 9: Contouring

Take spot levels and prepare a contour map for a given area.

> [illegible]

## Module 4: Theodolite Survey

### Expt 10: Repetition Method

Measure horizontal angles with high precision using the repetition method with a theodolite.

### Expt 12: Prolonging a Line

Prolong a straight line and measure the bearing of a line using a theodolite.

> [illegible]

## Practice Model Question Paper

*Practice Model Paper — not an official REV2026 paper*

1. Determine the level difference between two given points using the Height of Instrument (HI) method.

2. Measure the horizontal angle between two objects using the repetition method with a theodolite.

# Source Declaration

**Curriculum Scheme:** Revision 2026

**Subject Name:** Basic Surveying Lab

**Subject Code:** 2019

**Official Source URL:** SITTTR Link

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> This handbook was generated only from the identified scheme-specific SITTTR source. Content from other curriculum revisions was not merged.